Low Power Configurable Multi Clock System

Gate-level simulation: waveform snippets

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About this report

This short report shows selected snippets of the gate-level simulation waveform of System_Top (testbench entity *System_Top_tb*, captured in Questa Sim from the file GLS_wave.ps). The time axis is in picoseconds. Each snippet is a crop of the original waveform, recolored for print, with the decoded bytes and the signals to look at described underneath. Time windows of the capture that contain no traces are not shown.

Signals in the capture

Group	Signals	Role
Command path	RX_D_VALID, RX_P_DATA, Sync_RX_D_VALID, Sync_RX_P_DATA	received byte and valid flag, before and after the synchronizer into REF_CLK
Register file	RF_WrEn, RF_RdEn, RF_Rd_Data_Valid, RF_ADDR, RF_Wr_Data, RF_Rd_Data	strobes, address and data between SYS_CTRL and the register file
ALU	ALU_OP_A, ALU_OP_B, ALU_FUN, ALU_OUT, ALU_OUT_VALID	operands (REG0/REG1), function, result and valid
FIFO and TX	FIFO_WR_INC, FIFO_RD_INC, FIFO_EMPTY, FIFO_FULL, FIFO_WR_DATA, FIFO_RD_DATA, TX_BUSY, w_addr, r_addr, sync_w_ptr, sync_r_ptr, grey_w_ptr, grey_r_ptr	response path and the pointers that cross between clock domains
Configuration	CONFIG = 0x81, RX_CLK_DIV_RATIO = 1, TX_CLK_DIV_RATIO = 0x20	reset defaults: prescale 32, even parity on, TX divide 32

Snippets at a glance

#	Window	What it shows
1	0 – 1.2 ms	start-up register writes (aa 03 20, aa 02 81) and the first register read (bb 02 → 0x81), with the FIFO pointers
2	1.2 – 2.4 ms	read-back of register 3 (0x20); register writes to 0x04 (0x85) and 0x07 (0x1C)
3	3.2 – 4.4 ms	read-back sweep: 0x07 → 0x1C, 0x0B → 0xFF, 0x0F → 0x21
4	4.4 – 4.7 ms	ALU with operands: cc c8 07 00 → 0xCF
5	15.2 – 16.4 ms	operand preload (REG0 = 0x7A, REG1 = 0x0C) and the FIFO pointers for two queued response bytes

1. Start-up configuration and first register read

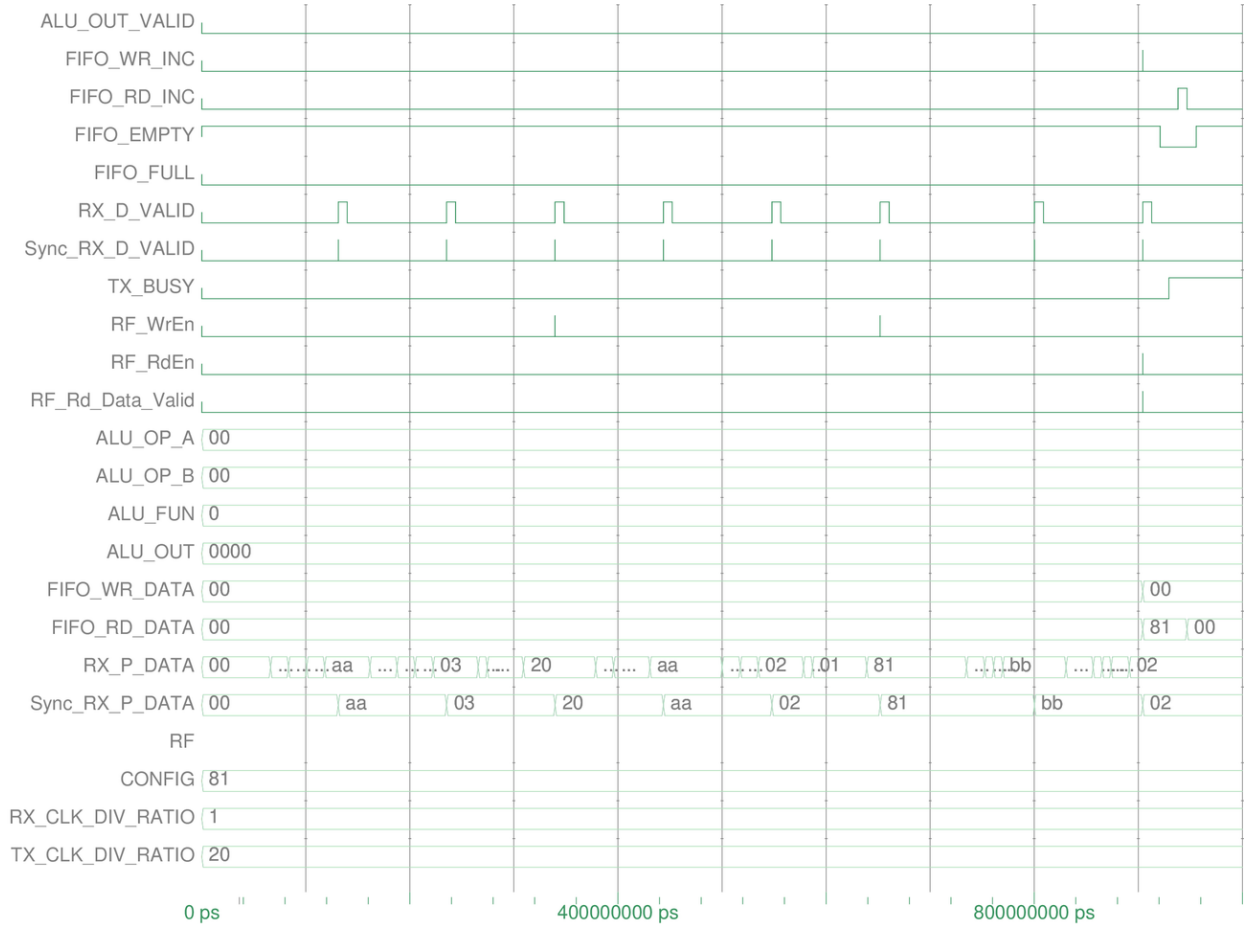


Figure 1. *Snippet 1: start-up writes and the first read (0 – 1.2 ms).*

Three commands arrive: **aa 03 20** writes 0x20 to register 3 and **aa 02 81** writes 0x81 to register 2 (one RF_WrEn pulse each, no response). **bb 02** reads register 2: RF_RdEn and RF_Rd_Data_Valid pulse, the byte is pushed to the FIFO (FIFO_WR_INC), FIFO_EMPTY falls in the read domain, TX_BUSY rises and FIFO_RD_INC pops the entry; FIFO_RD_DATA shows 0x81 and then 0x00.

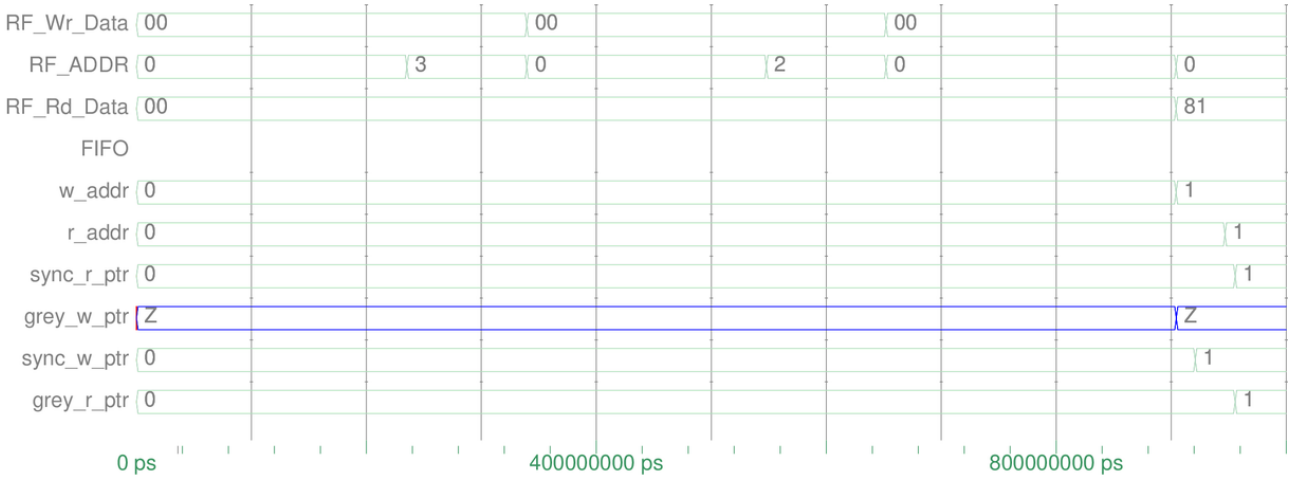


Figure 2. *Snippet 1 (same window): register-file address and FIFO pointers.*

RF_ADDR takes the values 3 and 2 for the two writes and returns to 0; RF_Rd_Data presents 0x81 on the read. After the write w_addr advances 0 → 1, the write pointer then reaches the read side (sync_w_ptr), and r_addr, grey_r_ptr and sync_r_ptr follow once the entry is popped. grey_w_ptr is displayed as Z in this netlist.

2. Register 3 read-back and register writes

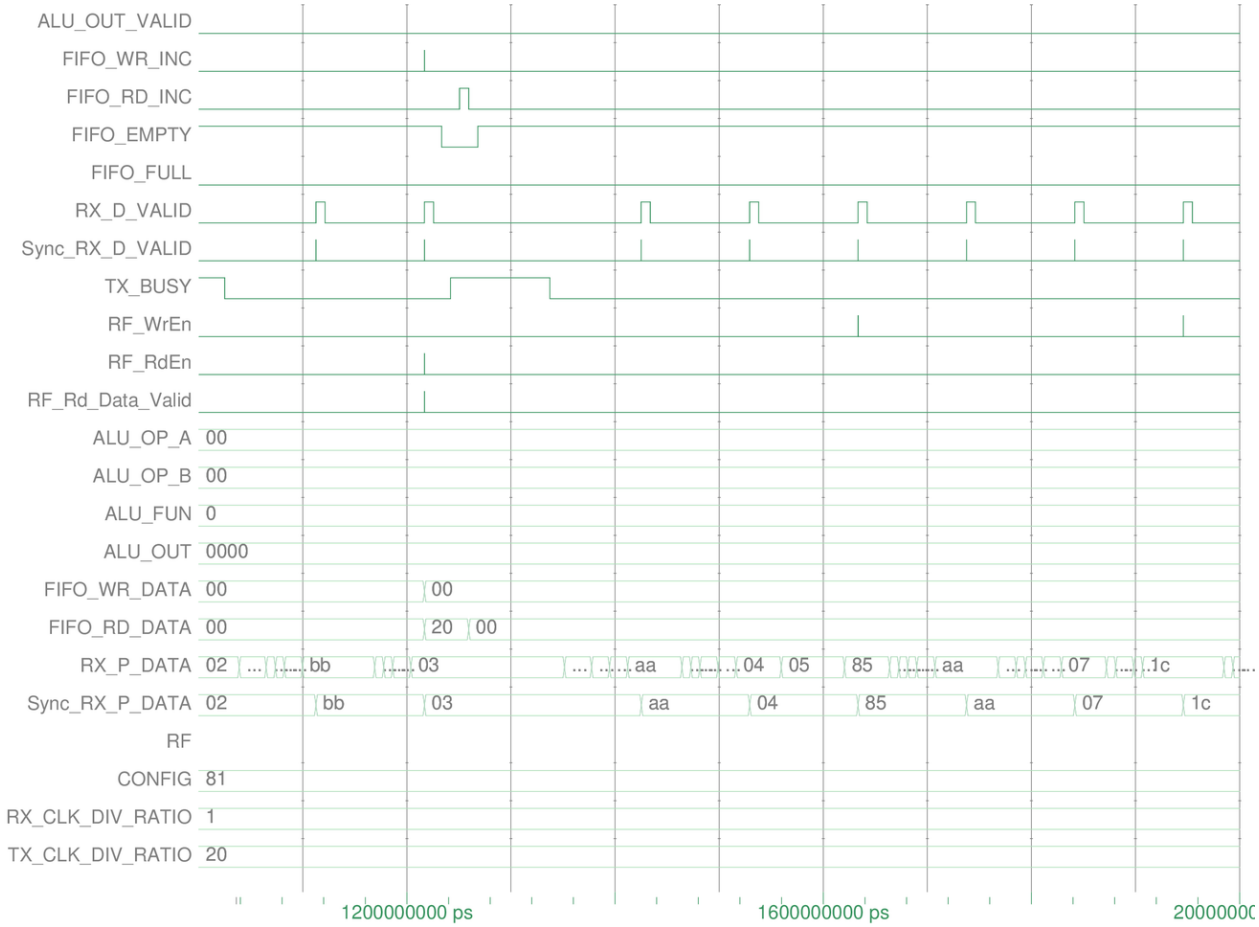


Figure 3. *Snippet 2: read of register 3 and writes to registers 0x04 and 0x07 (1.2 – 2.4 ms).*

bb 03 returns 0x20: it passes through the FIFO (FIFO_RD_DATA = 0x20, then 0x00 after the pop) and is transmitted (TX_BUSY). The two writes **aa 04 85** and **aa 07 1c** each produce one RF_WrEn pulse and no FIFO activity, because a write has no response.

3. Register read-back sweep

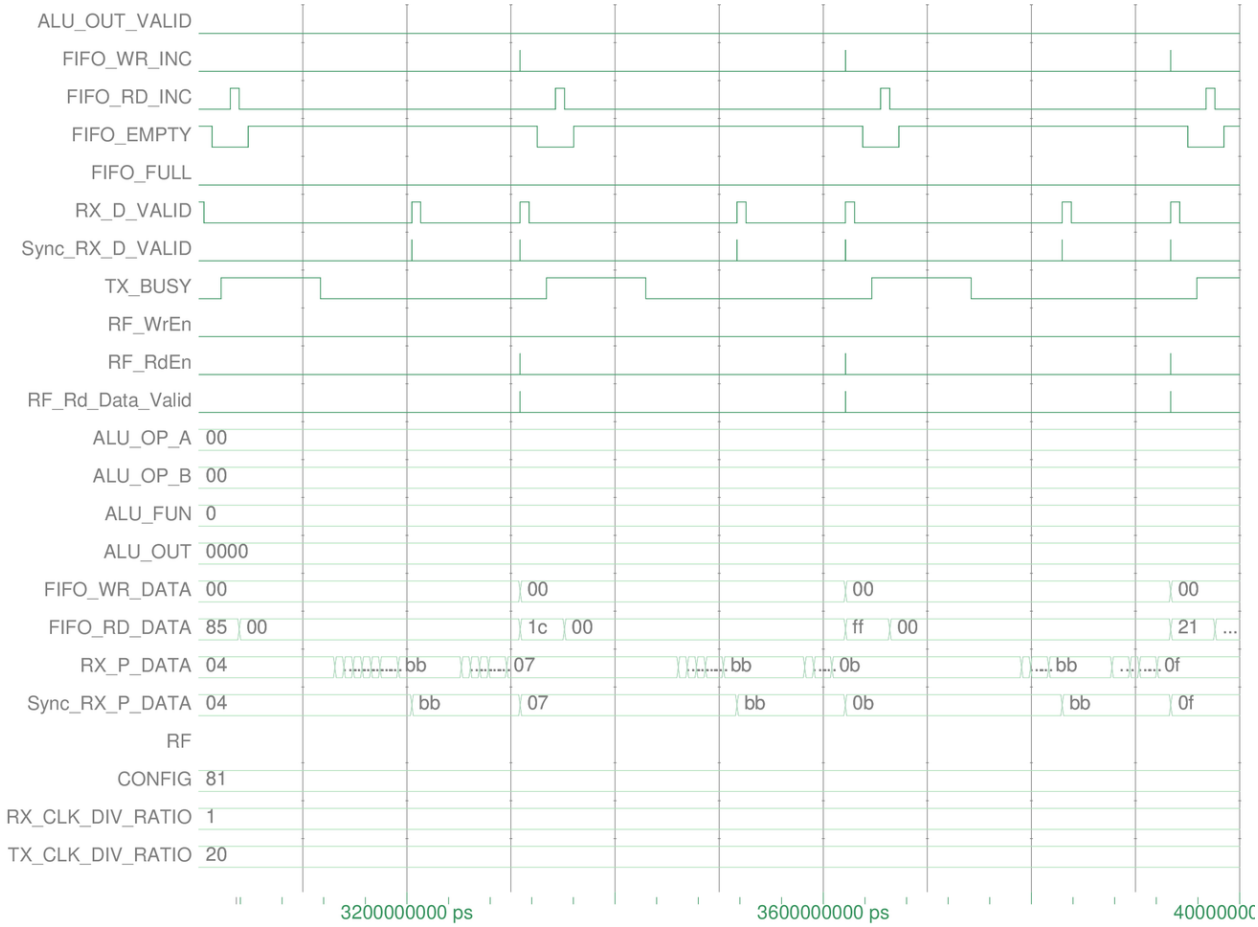


Figure 4. *Snippet 3: reads of registers 0x07, 0x0B and 0x0F (3.2 – 4.4 ms).*

Each **bb** command raises RF_RdEn and RF_Rd_Data_Valid, then FIFO_EMPTY falls, TX_BUSY rises and FIFO_RD_INC pops the byte. FIFO_RD_DATA shows the values returned: 0x85 (response of the previous read, at the left), then **0x1C**, **0xFF** and **0x21**, matching what was written earlier.

4. ALU operation with operands

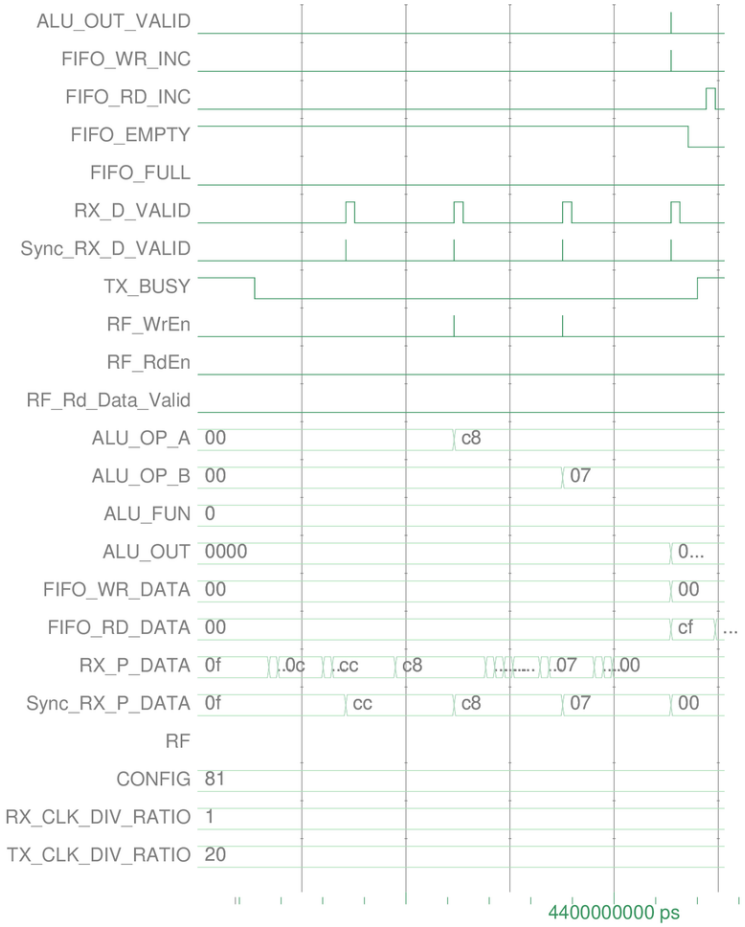


Figure 5. *Snippet 4: cc c8 07 00, ADD (4.4 – 4.7 ms).*

The frames **cc**, **c8**, **07**, **00** are decoded. Two RF_WrEn pulses load ALU_OP_A = 0xC8 and ALU_OP_B = 0x07 and ALU_FUN = 0 selects addition. ALU_OUT_VALID pulses together with FIFO_WR_INC, and FIFO_RD_DATA shows **0xCF** = 0xC8 + 0x07.

5. Operand preload and FIFO pointers

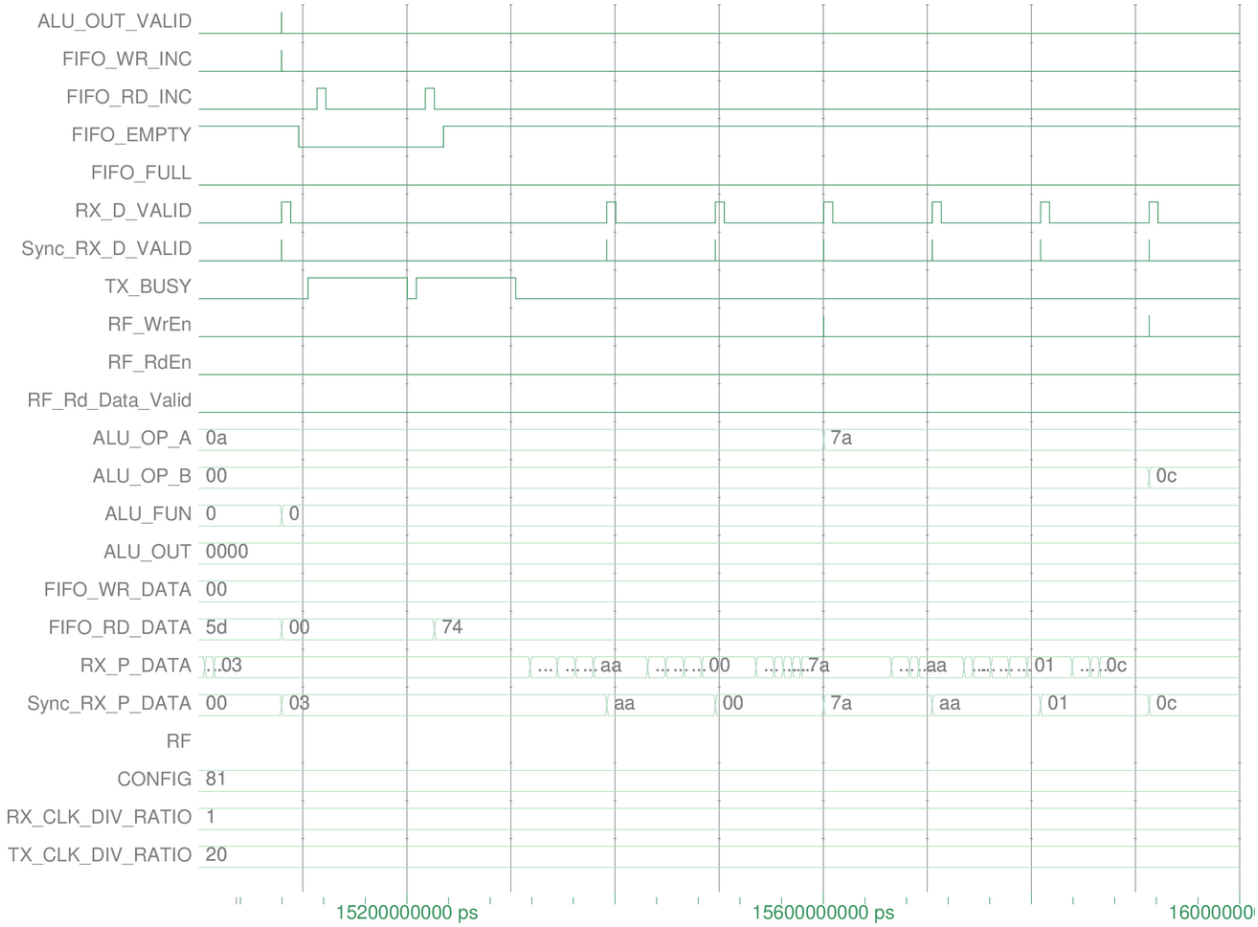


Figure 6. *Snippet 5: aa 00 7a and aa 01 0c (15.2 – 16.4 ms).*

REG0 = 0x7A and REG1 = 0x0C are written (ALU_OP_A and ALU_OP_B update, one RF_WrEn pulse each); these are the operands for the no-operand ALU command. At the left the FIFO is still draining the previous response bytes (FIFO_RD_DATA 0x5D, 0x00, 0x74) while TX_BUSY toggles.



Figure 7. *Snippet 5 (same window): FIFO pointers.*

w_addr advances 0 → 2 as two response bytes are written and r_addr follows 1 → 2 as they are popped. sync_w_ptr moves c → f (pointer 8 → 10 in gray code) after both writes, and grey_r_ptr and sync_r_ptr step c → d → f with the reads.

Observations

- All decoded values are as expected: 0x81 and 0x20 (reset defaults read back), 0x85, 0x1C, 0xFF and 0x21 (written then read back), and 0xCF for 0xC8 + 0x07.
- No unknown (X) values appear on the functional signals in the snippets shown, and FIFO_FULL stays low.
- grey_w_ptr is displayed as **Z** throughout: that net is not driven under this name in the netlist, which is probably a synthesis renaming. The crossing itself is visible through w_addr, sync_w_ptr, r_addr, grey_r_ptr and sync_r_ptr, which all update correctly.

6. Gate-level simulation log

Tests from the QuestaSim transcript of the run behind the waveforms above (SDF-annotated netlist, prescale 32, even parity).

✓ No setup / hold violations

No setup, hold, recovery, removal or width violation messages appear in the log from 0 ps to the end of the run (19.489 ms). All **61 tests passed, 0 failed**. SDF back-annotation completed successfully (max delays, `-sdfmax`).

```
   703.8 us  PASS  RX_CLK period 271.3 ns
   712.5 us  PASS  TX_CLK period 8680.6 ns
  1025.1 us  PASS  RD ADD A=0x02 B=0x00 LSB : 0x81
  1337.6 us  PASS  RD ADD A=0x03 B=0x00 LSB : 0x20
  2804.6 us  PASS  RF writes: no response
  3117.1 us  PASS  RD ADD A=0x04 B=0x00 LSB : 0x85
  3429.6 us  PASS  RD ADD A=0x07 B=0x00 LSB : 0x1c
  3742.1 us  PASS  RD ADD A=0x0b B=0x00 LSB : 0xff
  4054.6 us  PASS  RD ADD A=0x0f B=0x00 LSB : 0x21
  4575.5 us  PASS  CC ADD A=0xc8 B=0x07 LSB : 0xcf
  4679.6 us  PASS  CC ADD A=0xc8 B=0x07 MSB : 0x00
  5200.5 us  PASS  CC SUB A=0xc7 B=0x08 LSB : 0xbf
  5304.6 us  PASS  CC SUB A=0xc7 B=0x08 MSB : 0x00
  5825.5 us  PASS  CC MUL A=0xc6 B=0x09 LSB : 0xf6
  5929.6 us  PASS  CC MUL A=0xc6 B=0x09 MSB : 0x06
  6450.5 us  PASS  CC DIV A=0xc5 B=0x0a LSB : 0x13
  6554.6 us  PASS  CC DIV A=0xc5 B=0x0a MSB : 0x00
  7075.5 us  PASS  CC AND A=0xc4 B=0x0b LSB : 0x00
  7179.6 us  PASS  CC AND A=0xc4 B=0x0b MSB : 0x00
  7700.5 us  PASS  CC OR A=0xc3 B=0x0c LSB : 0xcf
  7804.7 us  PASS  CC OR A=0xc3 B=0x0c MSB : 0x00
  8325.5 us  PASS  CC NAND A=0xc2 B=0x0d LSB : 0xff
  8429.7 us  PASS  CC NAND A=0xc2 B=0x0d MSB : 0xff
  8950.5 us  PASS  CC NOR A=0xc1 B=0x0e LSB : 0x30
  9054.7 us  PASS  CC NOR A=0xc1 B=0x0e MSB : 0xff
  9575.5 us  PASS  CC XOR A=0xc0 B=0x0f LSB : 0xcf
  9679.7 us  PASS  CC XOR A=0xc0 B=0x0f MSB : 0x00
 10200.5 us  PASS  CC XNOR A=0xbf B=0x10 LSB : 0x50
 10304.7 us  PASS  CC XNOR A=0xbf B=0x10 MSB : 0xff
 10825.5 us  PASS  CC EQ A=0xbe B=0x11 LSB : 0x00
 10929.7 us  PASS  CC EQ A=0xbe B=0x11 MSB : 0x00
 11450.5 us  PASS  CC GT A=0xbd B=0x12 LSB : 0x02
 11554.7 us  PASS  CC GT A=0xbd B=0x12 MSB : 0x00
 12075.5 us  PASS  CC LT A=0xbc B=0x13 LSB : 0x00
 12179.7 us  PASS  CC LT A=0xbc B=0x13 MSB : 0x00
 12700.5 us  PASS  CC SHR A=0xbb B=0x14 LSB : 0x5d
 12804.7 us  PASS  CC SHR A=0xbb B=0x14 MSB : 0x00
 13325.5 us  PASS  CC SHL A=0xba B=0x15 LSB : 0x74
 13429.7 us  PASS  CC SHL A=0xba B=0x15 MSB : 0x01
 13950.5 us  PASS  CC EQ A=0x37 B=0x37 LSB : 0x01
 14054.7 us  PASS  CC EQ A=0x37 B=0x37 MSB : 0x00
 14575.5 us  PASS  CC MUL A=0xff B=0xff LSB : 0x01
 14679.7 us  PASS  CC MUL A=0xff B=0xff MSB : 0xfe
 15200.5 us  PASS  CC DIV A=0x0a B=0x00 LSB : 0x00
 15304.7 us  PASS  CC DIV A=0x0a B=0x00 MSB : 0x00
 16146.7 us  PASS  REG0/REG1 writes: no resp
 16459.2 us  PASS  DD ADD A=0x7a B=0x0c LSB : 0x86
 16563.4 us  PASS  DD ADD A=0x7a B=0x0c MSB : 0x00
 16875.9 us  PASS  DD SUB A=0x7a B=0x0c LSB : 0x6e
 16980.0 us  PASS  DD SUB A=0x7a B=0x0c MSB : 0x00
 17292.5 us  PASS  DD AND A=0x7a B=0x0c LSB : 0x08
 17396.7 us  PASS  DD AND A=0x7a B=0x0c MSB : 0x00
 17709.2 us  PASS  DD SHL A=0x7a B=0x0c LSB : 0xf4
 17813.4 us  PASS  DD SHL A=0x7a B=0x0c MSB : 0x00
 18125.9 us  PASS  DD GT A=0x7a B=0x0c LSB : 0x02
 18230.0 us  PASS  DD GT A=0x7a B=0x0c MSB : 0x00
 18750.9 us  PASS  CC DIV A=0x21 B=0x03 LSB : 0x0b
 18855.0 us  PASS  CC DIV A=0x21 B=0x03 MSB : 0x00
 19167.5 us  PASS  DD MUL A=0x21 B=0x03 LSB : 0x63
 19271.7 us  PASS  DD MUL A=0x21 B=0x03 MSB : 0x00
 19488.7 us  PASS  end of sweep: TX idle
```

```
PASSED: 61 FAILED: 0 — ALL TESTS PASSED
```